

ALPACARUNS

Autonomous MoE Multi-Agent Trading on Alpaca

Go · Google ADK · Alpaca Trading API · MCP

A trading bot that is auditable end-to-end —
five factor voices, one risk validator, one journal.

Paper-only by default. Fail-closed. No silent trades.

Hackathon submission · Sept 2026

PROBLEM

Trading bots should never be opaque.

- Most retail "AI trading" demos hide what they actually decided and why.
- LLM-driven agents fail silently: malformed tool calls, missing prices, drifted state.
- Alpaca's free tier blocks options snapshots — naive bots break; smart bots degrade.
- Owners deserve one journal that explains every fill, HOLD, and halt.

Alpacaruns is built so every order is reproducible and explainable.

WHAT IT DOES

Two execution paths. One shared risk gate.

LLM path (cycle / monitor)

- SequentialAgent: Market → Analysis(Technical □ Sentiment) → TradeIdea → Risk → Execution.
- Each Alpaca call is a tool from the official MCP server — no hand-rolled REST.
- Supervises live decisions; human approval unless MODE=autonomous AND confidence clears.

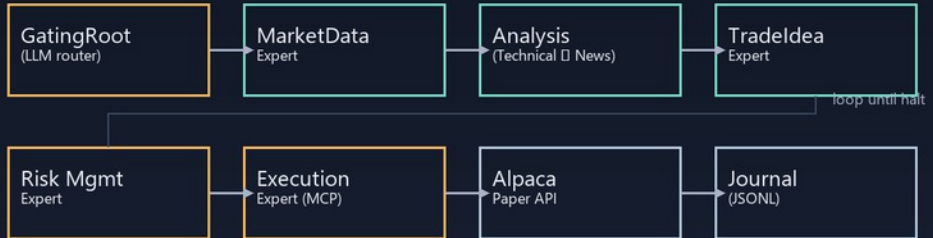
Deterministic path (auto)

- Pure pipeline: bars → five factors → threshold rule → size → risk → order.
- Identical inputs always produce identical outputs. No LLM, no surprise.
- Optional ensemble: six expert voices with performance-weighted gater.

Both paths funnel every order through the same Go-coded pre-trade gate.

ARCHITECTURE

Mixture-of-Experts graph on Google ADK for C



Six ADK experts. One shared validator. One JSONL journal.

RISK CONTROLS

Fail-closed. One validator. Every order path.

- Position notional cap (MAX_POSITION_USD) and portfolio-percentage cap (MAX_PORTFOLIO_PCT).
- Multi-factor gate: composite score $\geq$ FACTOR_MIN_SCORE; trend + momentum thresholds.
- Confidence floor: MIN_CONFIDENCE; ensemble needs $\geq$ 0.55 with a 2:1 directional split.
- Drawdown halts: daily 5%, weekly 10%, total 15% of peak — kill switch engages before any entry.
- Idempotent client_order_ids: no duplicate market-sells after a network timeout.

Hardening audit (Aug 26) verified all gates live; two money-path defects fixed.

TECH STACK

What we built it with.

Language

Go 1.22+

Agent graph

Google ADK for Go (Sequential / Parallel / Loop)

Broker

Alpaca Trading API via official MCP server (uvx alpaca-mcp-server)

LLM

Qwen3-4B-Instruct via llama.cpp (--jinja) · or Gemini · or gpt-oss-20b on Oxlo

State

JSONL journal + versioned strategy-state.json, temp+rename writes

Deploy

Distroless Docker + systemd unit, non-root, ProtectSystem=strict

One binary. No cloud account required to run paper.

HONEST CONSTRAINTS + DEMO

What we are not claiming. Live journal.

Honest constraints

- Paper-only so far — no live capital traded.
- Options overlay degrades to equity on free tier (404s).
- No backtester yet — thresholds validated by tests + paper runs.
- Two OPEN items: webhook notifier, journal rotation.

Live journal

```
buy TSLA qty=5 price=360.98  
tp=375.42 sl=353.76 crypto=false
```

```
ensemble-blocked BTC/USD conf=0.81  
votes=[trend:buy@1.00x0.75,  
breakout:buy@1.00x0.75,  
xsmom:buy@1.00x1.00,  
meanrev:hold@0.50x1.25]
```

```
# every decision journaled with  
# the full vote trail
```

alpacaruns auto --dry-run replays the same pipeline without orders.